Arista Access Node WFD-ACC-44 Optical Transport Failure Resolution SOP (ALARM_8504)

Node: WFD-ACC-44 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_8504

Description

The Optical Transport Failure (ALARM_8504) in the Access Node WFD-ACC-44, located at the Whitefield POP, is primarily caused by degradation of SFP/XFP modules, fiber cuts, or Rx/Tx power loss. This failure domain occurs due to the inherent limitations of optical transport protocols, such as Optical Transport Network (OTN) or Synchronous Digital Hierarchy (SDH), which are susceptible to signal degradation and power loss during transmission. In the context of the Arista Access Node, this failure is further exacerbated by the reliance on SFP/XFP modules, which are prone to degradation over time. Furthermore, the Optical Transport Network (OTN) protocol, implemented in the Access Node, is vulnerable to signal degradation and power loss during transmission, leading to the ALARM_8504. The failure of the SFP/XFP module or the fiber cut can be attributed to physical wear and tear, environmental factors, or manufacturing defects. To mitigate this failure, it is essential to perform regular inspections and maintenance activities on the SFP/XFP modules and the optical fibers to prevent signal degradation and power loss.

Pre-requisite Checks

- Verify the site location and node ID to ensure the correct node is being addressed.

- Confirm the vendor and role to ensure the correct CLI dialect and commands are being used.

- Check the redundant link status to ensure the network remains operational during the failure resolution process.

- Dispatch remote hands to the Whitefield POP to facilitate physical access to the node if necessary.

Potential Root Causes

Cause	Probability
SFP/XFP Module Degradation	High
Fiber Cut	Medium
Rx/Tx Power Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces fiber 1/1/1 detail, Expected Output: Port 1/1/1 is operational, and the SFP/XFP module is installed correctly., Description: This command isolates the optical transport protocol and the SFP/XFP module to determine if the issue is related to the module or the fiber cut.
```

```
Command: show optics fiber 1/1/1, Expected Output: The transmit and receive power levels are within the acceptable range., Description: This command checks the optical signal quality and power levels to determine if the issue is related to signal degradation or power loss.

Command: show log | grep ALARM_8504, Expected Output: The alarm code ALARM_8504 is present in the log output., Description: This command isolates the alarm code and logs to determine the root cause of the failure.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP Module, Command: replace module 1/1/1 with SFP-10G-SR, Description: This step resolves the issue by replacing the degraded SFP/XFP module with a new one, ensuring the optical transport protocol and signal quality are restored.

Step 2: Step: Repair or Replace the Fiber, Command: repair fiber 1/1/1 or replace fiber 1/1/1 with new fiber, Description: This step resolves the issue by repairing or replacing the fiber cut, ensuring the optical signal quality and power levels are restored.

Step 3: Step: Adjust the Rx/Tx Power Levels, Command: adjust optics fiber 1/1/1 to adjust the transmit and receive power levels, Description: This step resolves the issue by adjusting the optical signal quality and power levels to ensure the transmit and receive power levels are within the acceptable range.

Post-Check Verification

```
Command: show interfaces fiber 1/1/1 detail, Expected Output: Port 1/1/1 is operational, and the SFP/XFP module is installed correctly., Description: This command verifies the optical transport protocol and SFP/XFP module are operational and correctly installed.

Command: show optics fiber 1/1/1, Expected Output: The transmit and receive power levels are within the acceptable range., Description: This command verifies the optical signal quality and power levels are within the acceptable range.

Command: show log | grep ALARM_8504, Expected Output: The alarm code ALARM_8504 is not present in the log output., Description: This command verifies the alarm code ALARM_8504 is no longer present in the log output, indicating the issue is resolved.
```

Roll-back Procedures

- Step: Restore the Original SFP/XFP Module, Command: replace module 1/1/1 with original SFP/XFP module, Description: This step rolls back the change by restoring the original SFP/XFP module, ensuring the optical transport protocol and signal quality are restored to their original state.

- Step: Restore the Original Fiber, Command: restore fiber 1/1/1 to original state, Description: This step rolls back the change by restoring the original fiber, ensuring the optical signal quality and power levels are restored to their original state.

- Step: Restore the Original Rx/Tx Power Levels, Command: restore optics fiber 1/1/1 to original state, Description: This step rolls back the change by restoring the original Rx/Tx power levels, ensuring the transmit and receive power levels are restored to their original state.